Machine Learning Exercise 3B

Problem 1 *In the previous exercise, you used logistic regression to classify Fashion MNIST images. However, logistic regression is a linear classifier and cannot capture complex patterns. In this exercise, you will implement neural networks using TensorFlow/Keras and explore how network architecture affects performance.*

You will use the same Fashion MNIST dataset as before, which contains 28×28 *grayscale images of clothing items in 10 categories.*

i) Load the Fashion MNIST dataset and preprocess it:

- *Normalize pixel values to* $[0, 1]$
- *For dense networks: flatten images to* 784*-dimensional vectors*
- *For CNN: reshape images to* $28 \times 28 \times 1$ *(adding channel dimension)*
- *Convert labels to one-hot encoding using* `to_categorical`
- *Split the training set into training (*50000 *samples) and validation (*10000 *samples)*

ii) Create a ***simple neural network*** *using the* `Sequential` *API with:*

- *Input layer:* 784 *units*
- *Hidden layer:* 32 *units with ReLU activation*
- *Output layer:* 10 *units with softmax activation*

iii) Compile the model using:

- *Optimizer:* `adam`
- *Loss function:* `categorical_crossentropy`
- *Metrics:* `accuracy`

iv) Train the simple model for 50 *epochs and record the training and validation accuracy.*

v) Create a ***deeper neural network*** *with:*

- *Input layer:* 784 *units*
- *Hidden layer 1:* 256 *units with ReLU activation*
- *Hidden layer 2:* 128 *units with ReLU activation*
- *Hidden layer 3:* 64 *units with ReLU activation*
- *Output layer:* 10 *units with softmax activation*

vi) Train the deeper model for 50 *epochs and record the training and validation accuracy.*

vii) Create a ***Convolutional Neural Network (CNN)*** *with:*

- *Input:* $28 \times 28 \times 1$
- *Conv2D:* 32 *filters,* 3×3 *kernel, ReLU activation*
- *MaxPooling2D:* 2×2
- *Conv2D:* 64 *filters,* 3×3 *kernel, ReLU activation*

- *MaxPooling2D:* 2×2
- *Flatten*
- *Dense:* 64 *units with ReLU activation*
- *Output:* 10 *units with softmax activation*

viii) Train the CNN for 50 *epochs and record the training and validation accuracy.*

ix) Plot the learning curves (training and validation accuracy vs. epochs) for all three models.

x) Create a summary table comparing all architectures:

- *Number of parameters*
- *Training accuracy*
- *Validation accuracy*
- *Test accuracy*
- *Training time*

xi) Evaluate all models on the test set. Answer the following questions:

- *Which architecture performs best?*
- *Why do CNNs perform better than fully-connected networks on image data?*
- *Compare the CNN with the Deep fully-connected network: which has more parameters? Which performs better?*
- *What is the accuracy improvement per parameter for each model?*